

Study and Verification of the operation of Half and Full Adder and Subtractor circuit.

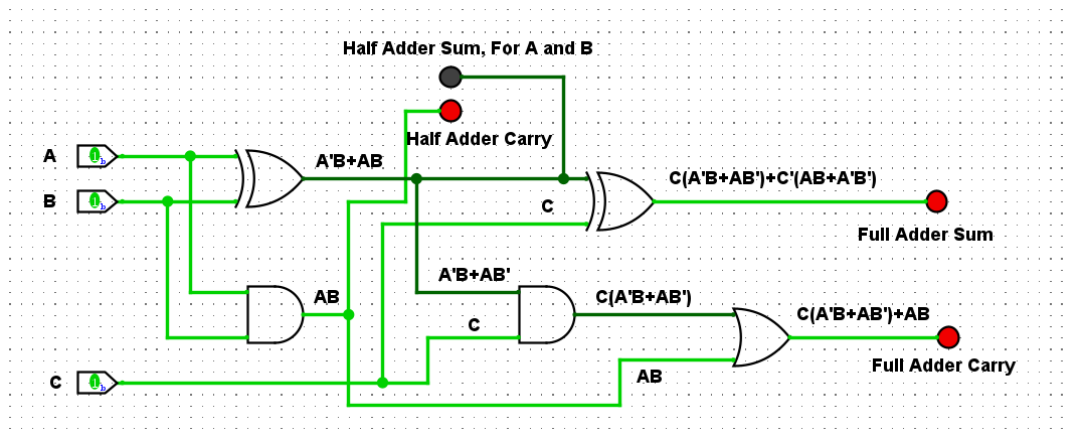


Figure 9: Half and Full Adder Circuit using Logic Gates

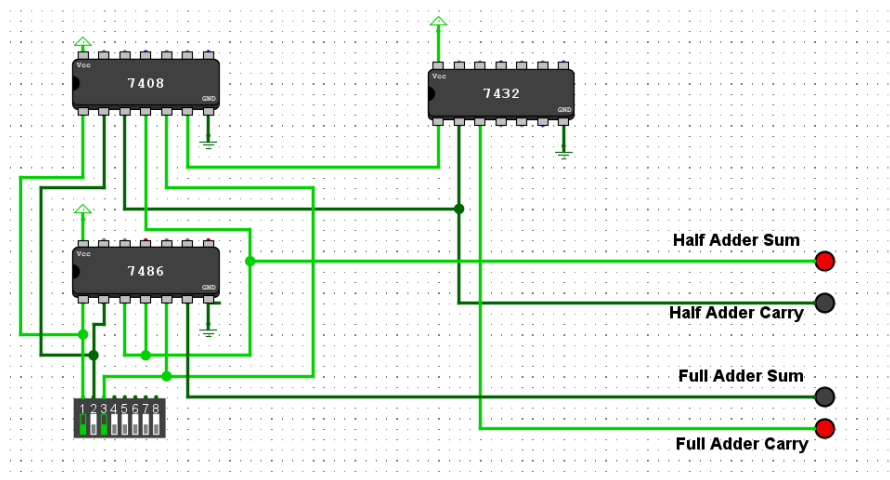


Figure 10: Half and Full Adder Circuit using Logic IC's

```

1 module adder (
2   input A,B,C,
3   output sum_half,carry_half,
4   output sum_full,carry_full
5 );
6
7 //Half Adder
8 assign sum_half = A ^ B; //XOR gate
9 assign carry_half = A & B; //AND gate
10
11 //Full Adder
12 assign sum_full = A ^ B ^ C; //XOR gate
13 assign carry_full = (A & B) | (B & C) | (C & A); //Carry out using AND and OR gates
14
15
16 endmodule

```

```

---- Half Adder ----
A B | sumH carryH
0 0 | 0 0
0 1 | 1 0
1 0 | 1 0
1 1 | 0 1

---- Full Adder ----
A B C | sumH carryH | sumF carryF
0 0 0 | 0 0 | 0 0
0 0 1 | 0 0 | 1 0
0 1 0 | 1 0 | 1 0
0 1 1 | 1 0 | 0 1
1 0 0 | 1 0 | 1 0
1 0 1 | 1 0 | 0 1
0 1 1 | 1 0 | 0 1
1 1 1 | 0 1 | 1 1
tb.v:77: $finish called at 300000 (1ps)

```

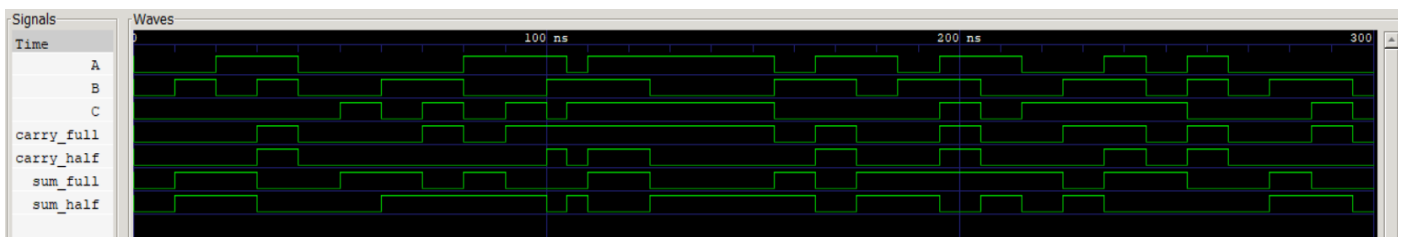


Figure 11: Verilog Code with corresponding Truth table and Signal Response for both Half Adder and Full Adder

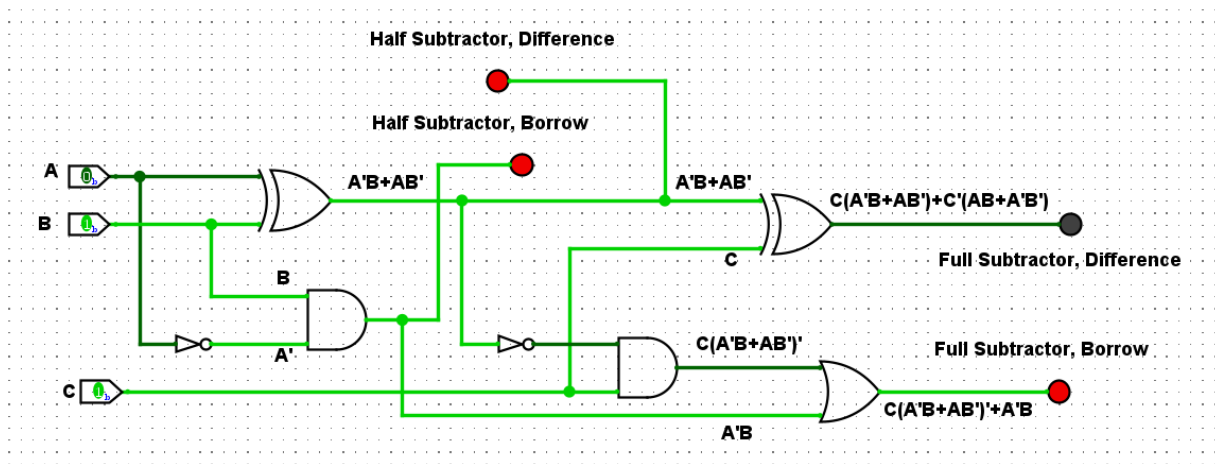


Figure 12: Half and Full Subtractor using Logic Gates

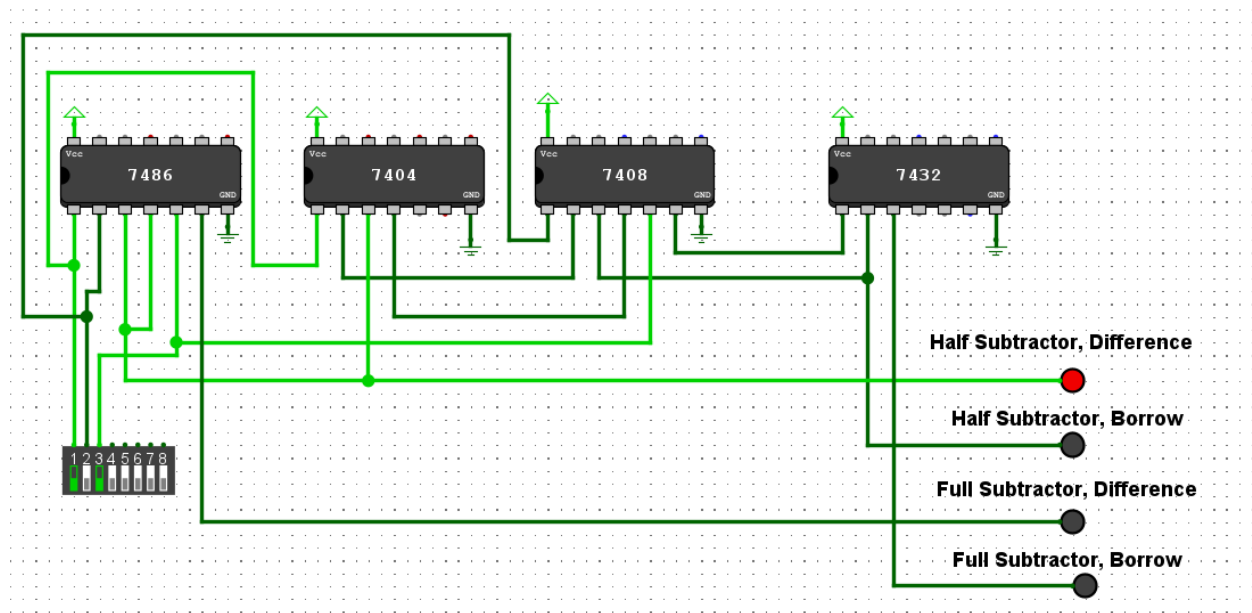


Figure 13: Half and Full Subtractor using IC's

```

1 module subtractor (
2     input A, B, C,          // Bin = Borrow in
3     output diff_half, borrow_half,
4     output diff_full, borrow_full
5 );
6
7     // Half Subtractor
8     assign diff_half = A ^ B;          // Difference
9     assign borrow_half = ~A & B;      // Borrow
10
11    // Full Subtractor
12    assign diff_full = A ^ B ^ C;      // Difference
13    assign borrow_full = (~A & B) | ((~A | B) & C); // Borrow out
14
15 endmodule

```

```

---- Half Subtractor ---
A B | diffH borrowH
0 0 | 0 0
0 1 | 1 1
1 0 | 1 0
1 1 | 0 0

---- Full Subtractor ----
A B C | diffH borrowH | diffF borrowF
0 0 0 | 0 0 | 0 0
0 0 1 | 0 0 | 1 1
0 1 0 | 1 1 | 1 1
0 1 1 | 1 1 | 0 1
1 0 0 | 1 0 | 1 0
1 0 1 | 1 0 | 0 0
0 1 1 | 1 1 | 0 1
1 1 1 | 0 0 | 1 1
tb.v:77: $finish called at 300000 (1ps)

```

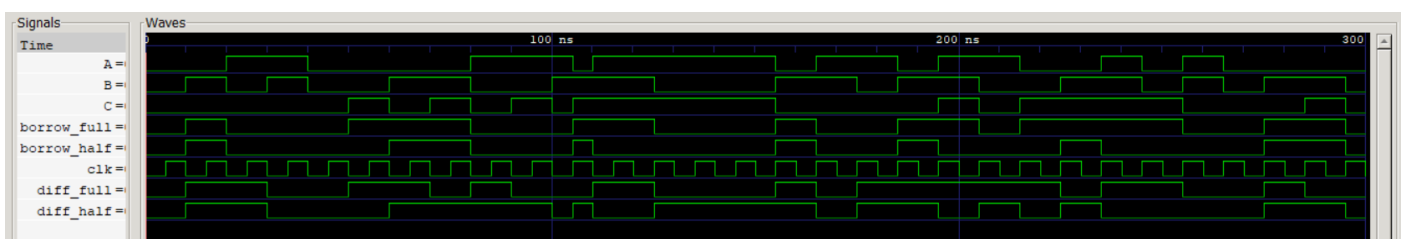


Figure 14: *Verilog Code with corresponding Truth table and Signal Response for both Half Subtractor and Full Subtractor*